THE NOVA RESEARCH CURRICULUM

Mathematics

Grade 8 · Term 2

Student edition

Outcomes: MA-8.2 simultaneous equations, linear inequalities, y = mx + c; MA-8.3 Pythagoras, arc length and sector area; MA-8.5 congruence, similarity, rotation and enlargement.

Lesson notes

- **MA-8.2 (Weeks 1–4):** solve pairs of equations by elimination, substitution and graphs (the crossing point). Linear inequalities on a number line (x > 3, open circle). Y = mx + c: m is the gradient, c the y-intercept.

  ▶ **Try it on NovaSchool:** Gradient and intercept: y = mx + c — open **Simulations** on your dashboard.

- **MA-8.3 (Weeks 5–8):** Pythagoras: in a right-angled triangle, $a^2 + b^2 = c^2$ (c is the hypotenuse); Pythagorean triples (3, 4, 5). Arc length = (θ/360) × 2πr; sector area = (θ/360) × πr^2.

  ▶ **Try it on NovaSchool:** Pythagoras: squares on the sides — open **Simulations** on your dashboard.

- **MA-8.5 (Weeks 9–12):** congruent shapes are the same shape and size (SSS, SAS, ASA, RHS); similar shapes have equal angles and sides in the same ratio. Rotation about a point (90°, 180°); enlargement with a scale factor from a centre.

Workbook pages

Page	Task
1–2	Solve 8 pairs of simultaneous equations; graph two.
3	Gradient and intercept from equations and graphs.
4	Pythagoras problems, including a ladder against a wall.
5	Arc and sector problems.
6	Rotation and enlargement on grid paper.

Question bank

#	Outcome	Question	Your answer
1	MA-8.2	Solve x + y = 10 and x − y = 4.	x = 6, y = 4 / x = 7, y = 3 / x = 5, y = 5 / x = 4, y = 6
2	MA-8.2	In y = 3x − 2, the gradient is …	−2 / 3 / 1 / x
3	MA-8.2	Solve 2x + 1 > 7.	x > 4 / x > 3 / x < 3 / x = 3
4	MA-8.3	A right-angled triangle has legs 6 cm and 8 cm. Hypotenuse?	14 cm / 12 cm / 10 cm / 48 cm

#	Outcome	Question	Your answer
5	MA-8.3	Which is a Pythagorean triple?	2, 3, 4 / 5, 12, 13 / 4, 5, 6 / 1, 2, 3
6	MA-8.3	Arc length of a 90° sector with r = 14 cm (π = 22/7)?	44 cm / 22 cm / 88 cm / 11 cm
7	MA-8.5	Similar shapes always have the same size. True or false.	True / False
8	MA-8.5	A 2 cm side enlarged by scale factor 3 becomes …	5 cm / 6 cm / 2/3 cm / 9 cm
9	MA-8.5	Which rule proves two right-angled triangles congruent using hypotenuse and one side?	SSS / ASA / RHS / AAA